

SHETH L.U.J. AND SIR M.V. COLLEGE
SUBJECT: Data Visualization with Power BI and Tableau

PRACTICAL NO. 8

AIM: Performing Advanced Analytics using Trends, Clustering, and Forecasting

a) Add and interpret trend lines (linear, exponential)

b) Customize and analyze trend models

c) Perform clustering on datasets

d) Analyze distributions using built-in tools

e) Apply forecasting techniques

f) Compare different statistical models

g) Export and interpret statistical results

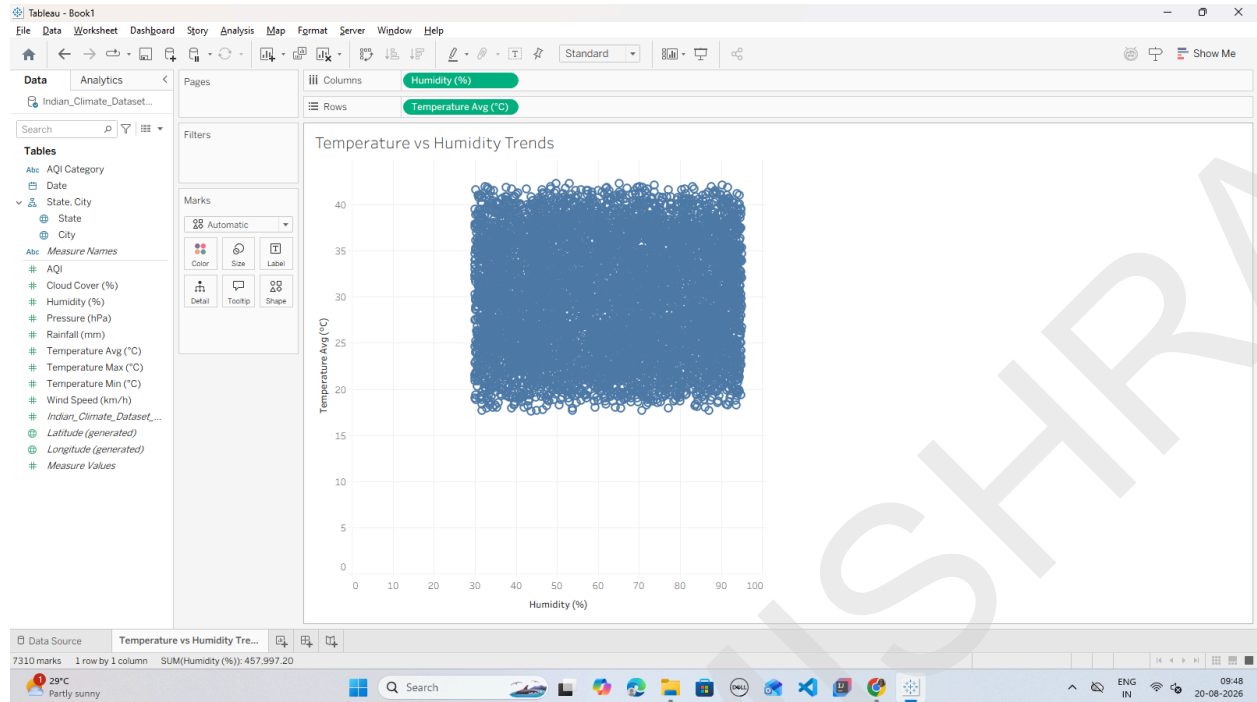
A: Add and Interpret Trend Lines (Linear, Exponential)

1. Connect Dataset & Build a Scatter Plot / Line Chart

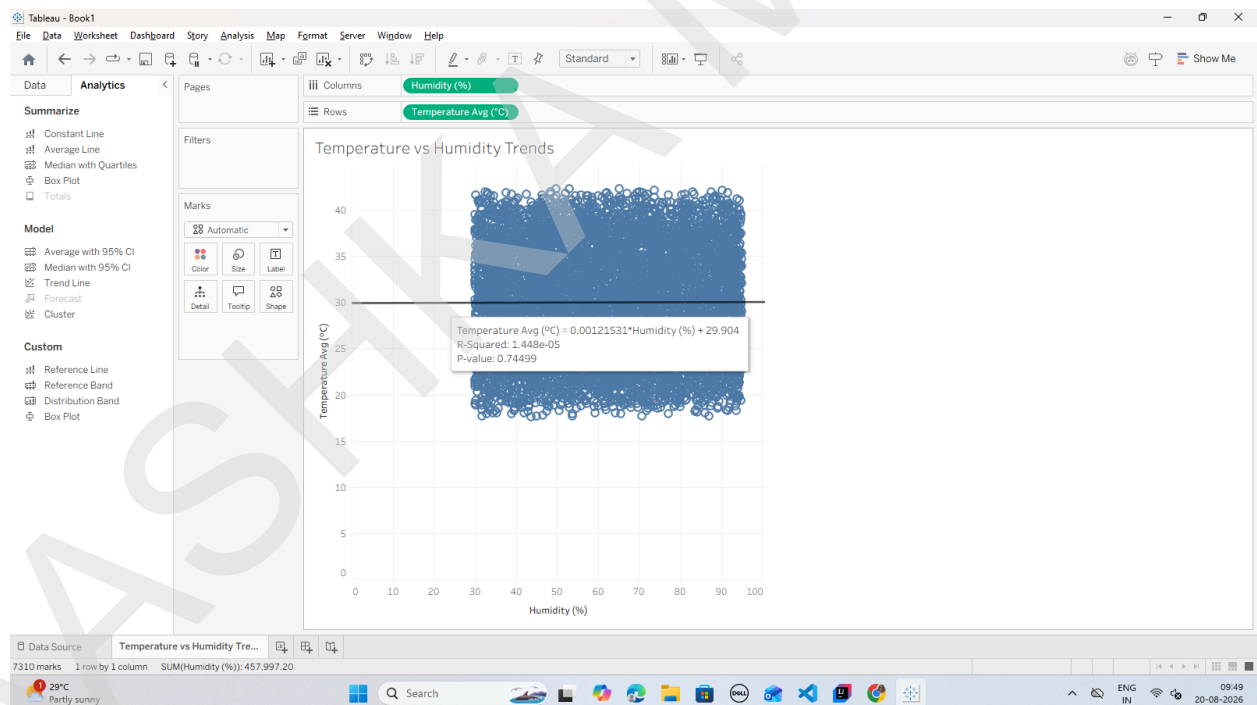
The screenshot displays the Tableau Desktop interface. On the left, the 'Connections' pane shows the 'Indian_Climate_Dataset_2024_2025' dataset connected. Below it, the 'Files' pane lists various CSV files. The main workspace shows the 'Indian_Climate_Dataset_2024_2025' dataset with 13 fields and 7310 rows. A table view of the dataset is displayed, showing columns for Date, City, State, and Temperature Max (°C). The table contains data for various cities and states, including Mumbai, Delhi, Bengaluru, Chennai, Kolkata, Hyderabad, Ahmedabad, Jaipur, Lucknow, Bhopal, and others.

Date	City	State	Temperature Max (°C)
01-01-2024	Mumbai	Maharashtra	32.5000
01-01-2024	Delhi	Delhi	25.4000
01-01-2024	Bengaluru	Karnataka	37.2000
01-01-2024	Chennai	Tamil Nadu	37.2000
01-01-2024	Kolkata	West Bengal	27.4000
01-01-2024	Hyderabad	Telangana	44.4000
01-01-2024	Ahmedabad	Gujarat	32.8000
01-01-2024	Jaipur	Rajasthan	40.4000
01-01-2024	Lucknow	Uttar Pradesh	42.3000
01-01-2024	Bhopal	Madhya Pradesh	27.4000
02-01-2024	Mumbai	Maharashtra	25.6000
02-01-2024	Delhi	Delhi	30.8000
02-01-2024	Bengaluru	Karnataka	41.1000
02-01-2024	Chennai	Tamil Nadu	33.3000

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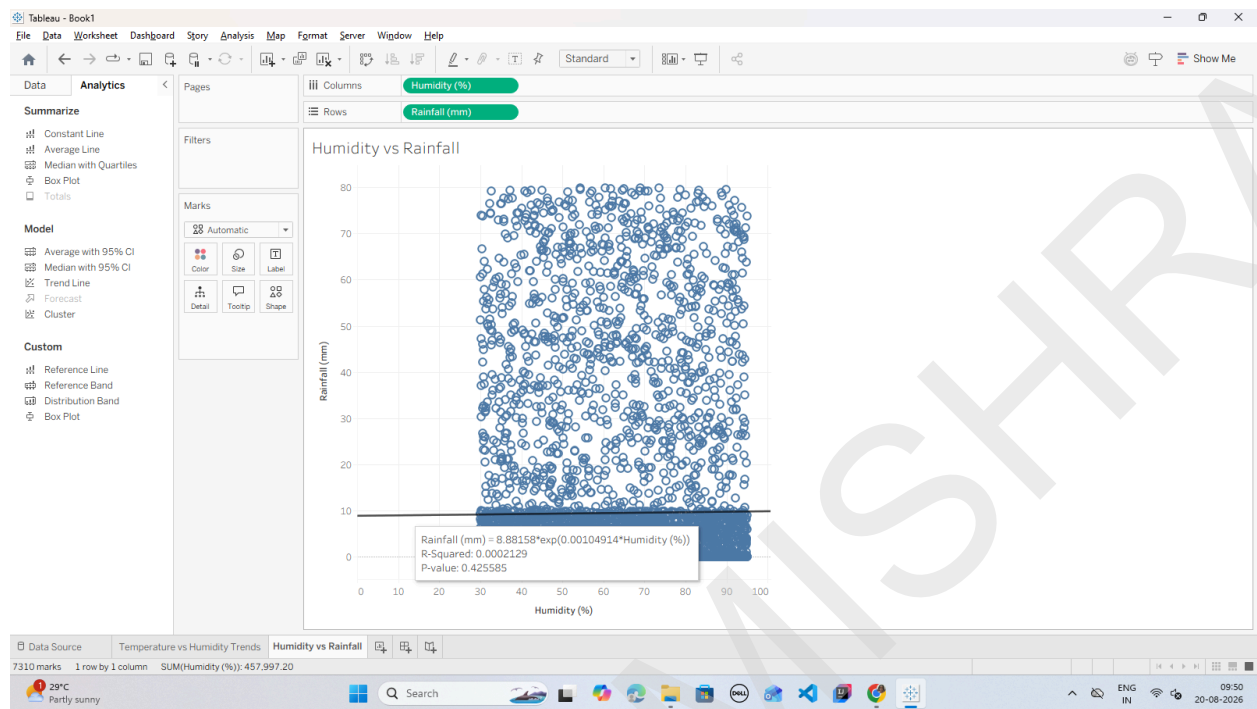
2. Add a Linear Trend Line



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ROLL NO: T095

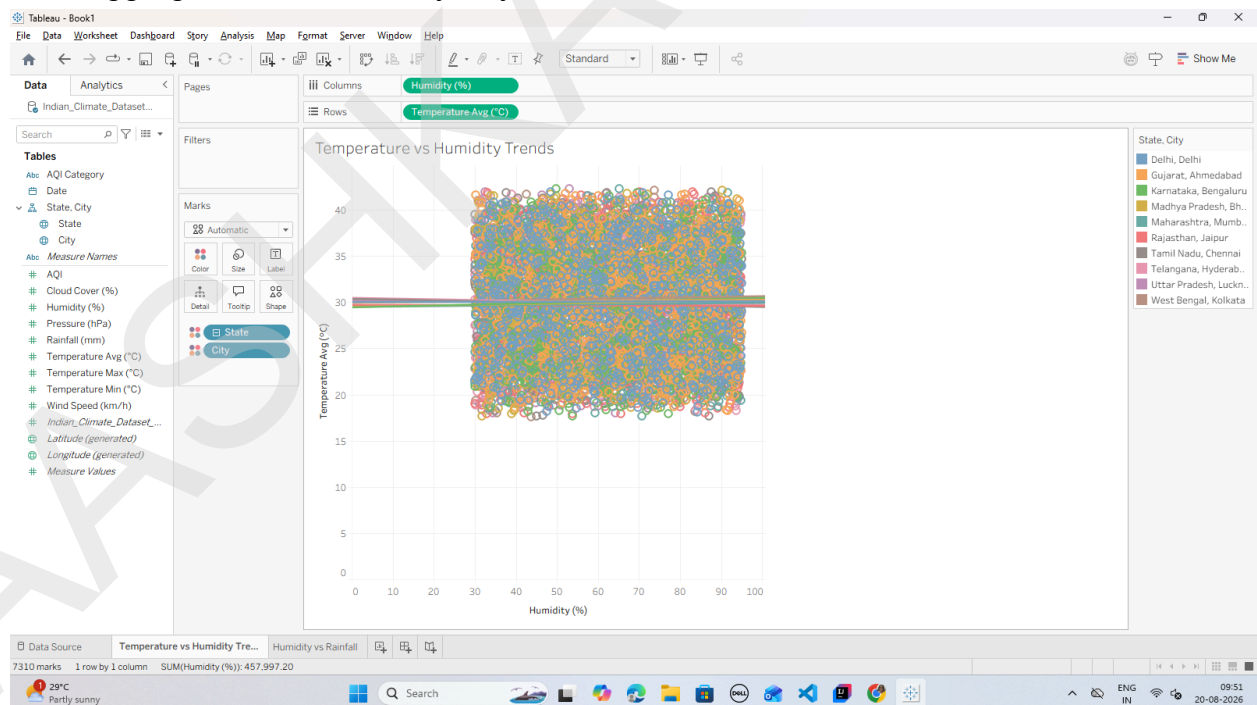
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3. Add an Exponential Trend Line for Non-Linear Measures (Rainfall vs. Humidity)



B: Customize and Analyze Trend Models

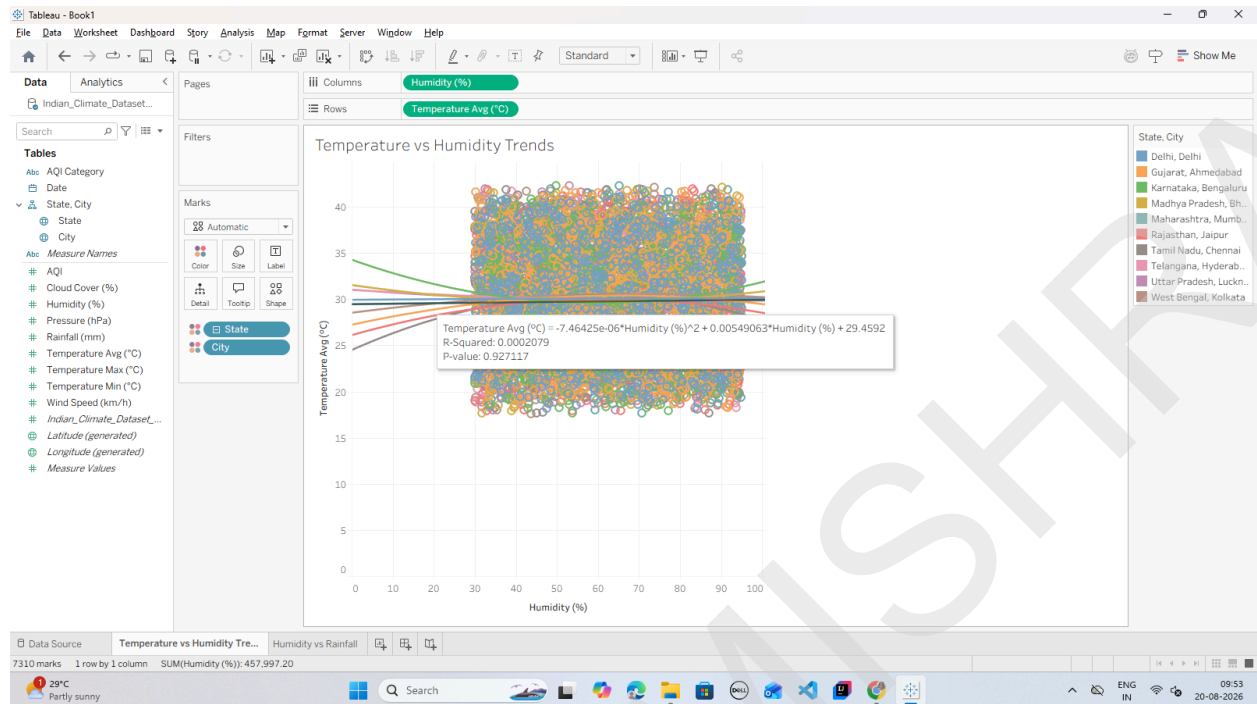
1. Disaggregate Trend Lines by City



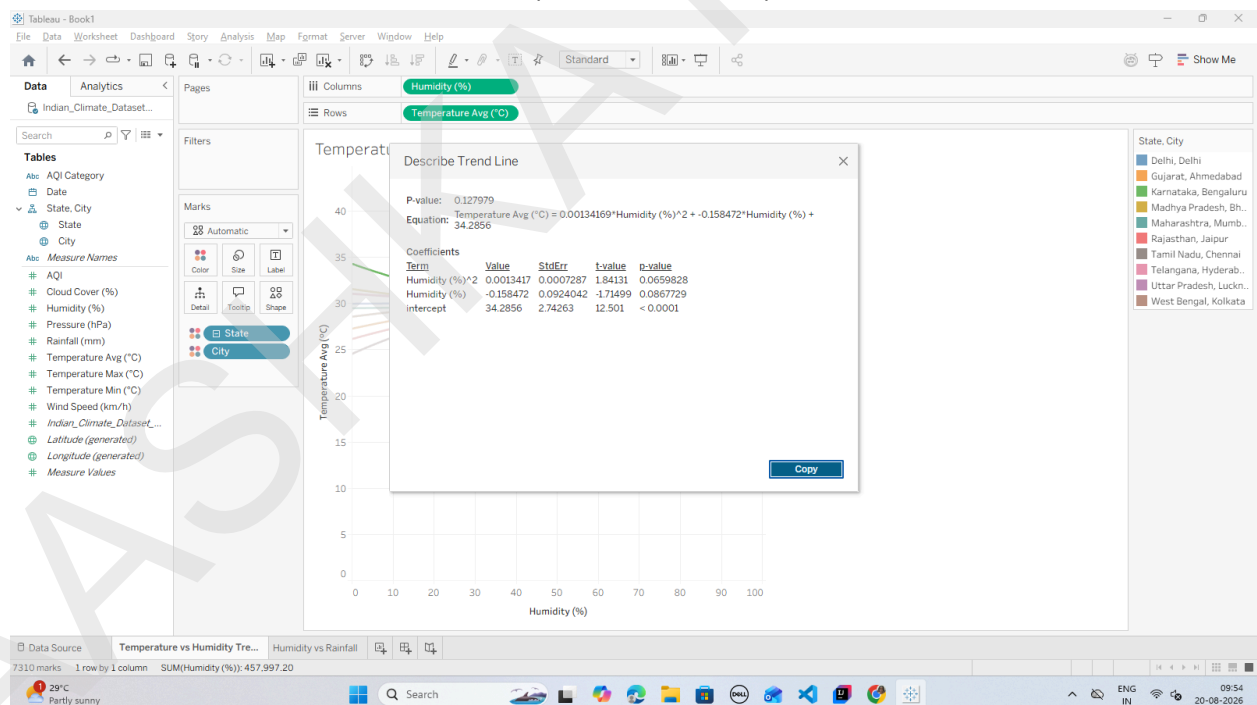
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2. Compare Different Model Types (Polynomial & Logarithmic)



3. Access Full Statistical Summaries (Describe Model)

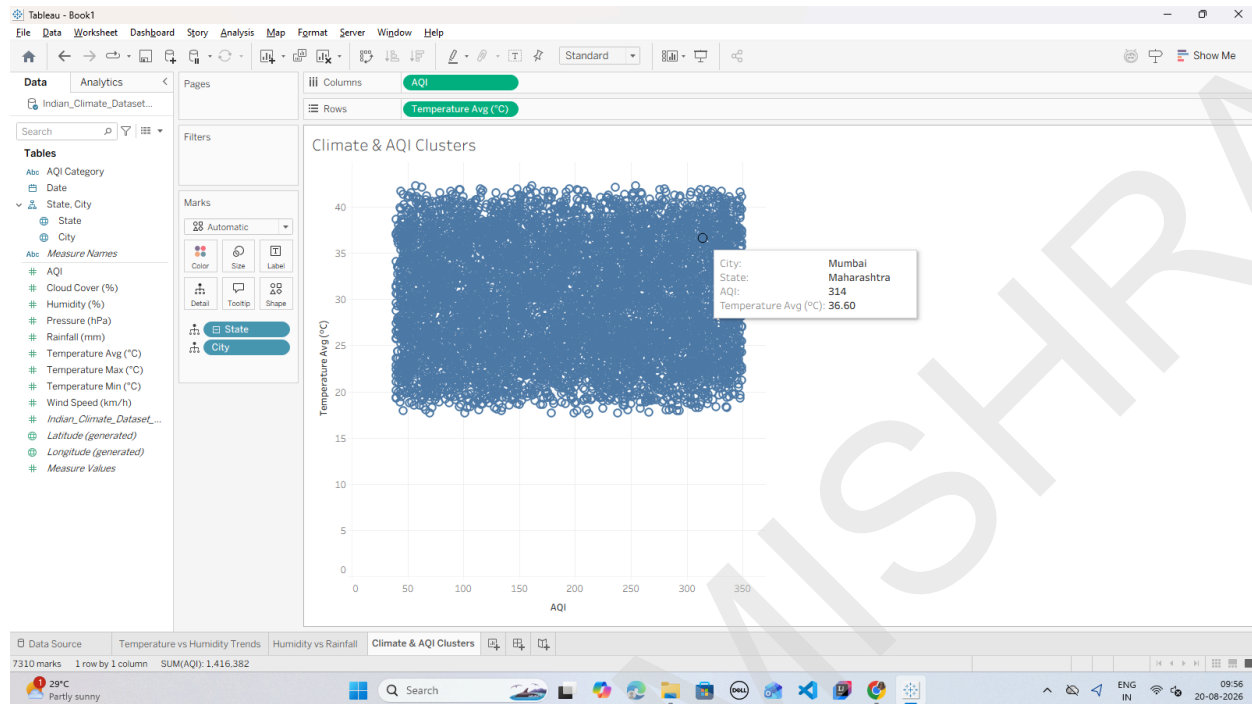


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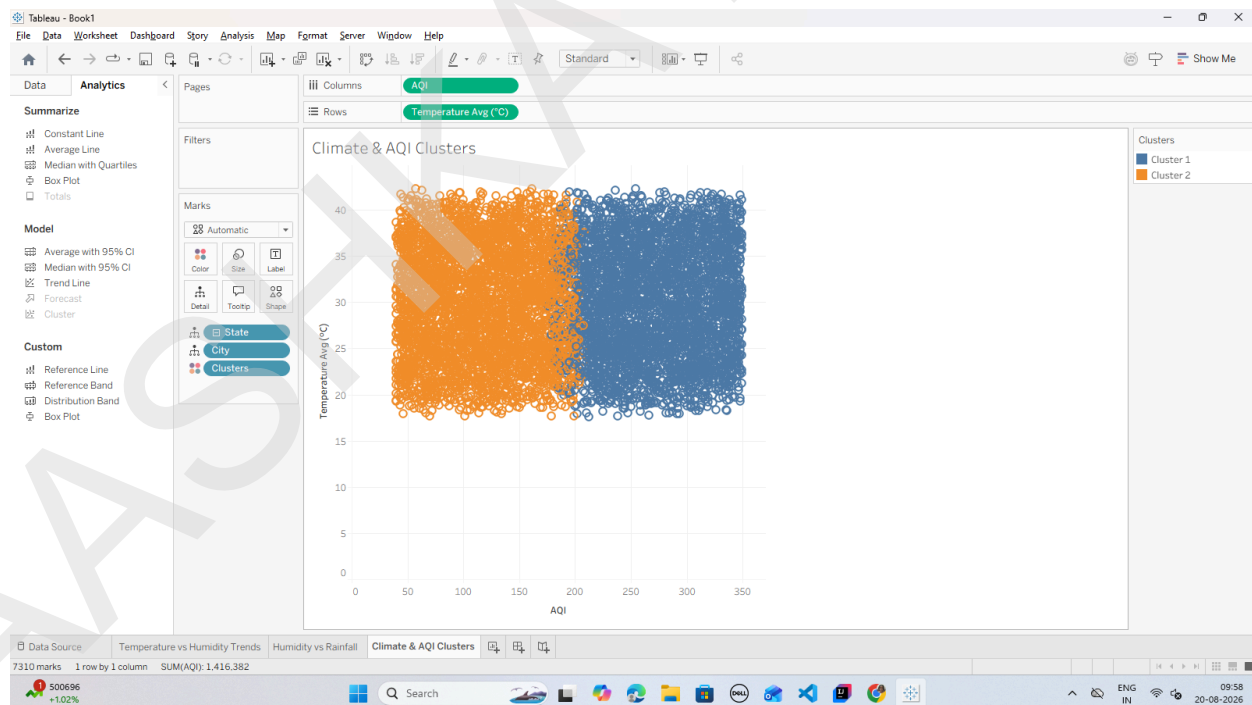
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C: Perform Clustering on Datasets

1. Build a Multi-Metric Scatter Plot Matrix

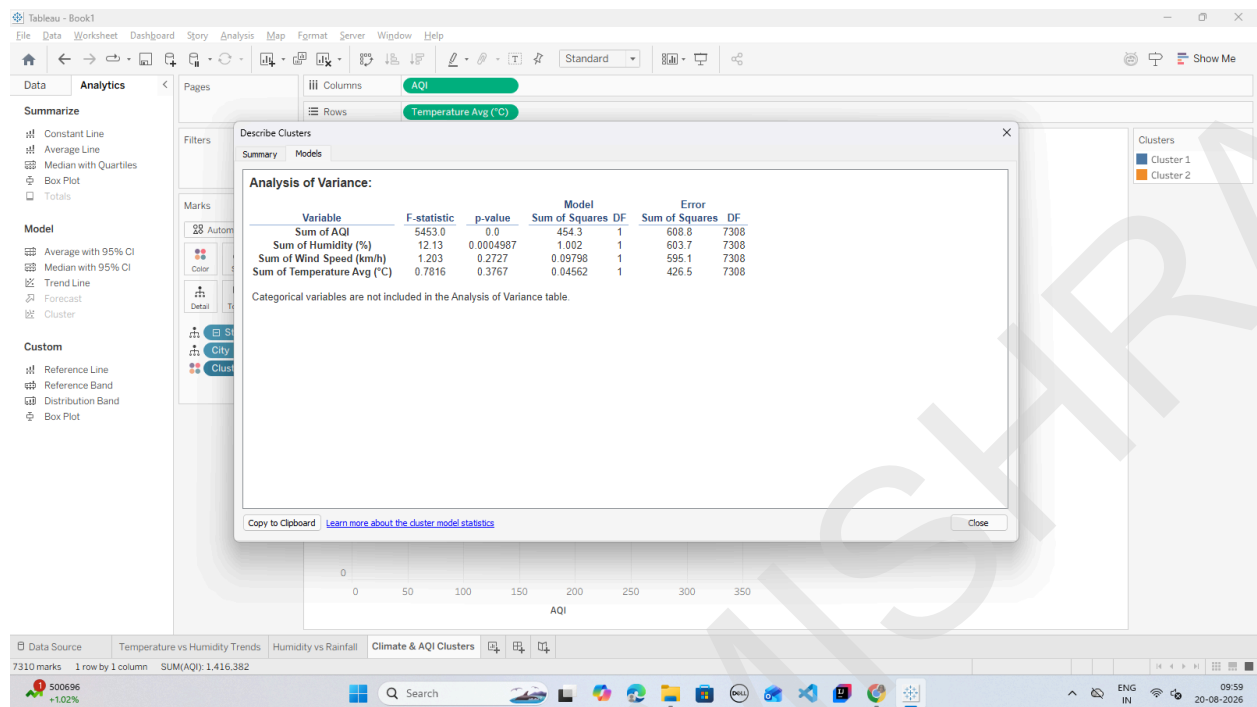


2. Apply Built-in Tableau Clustering



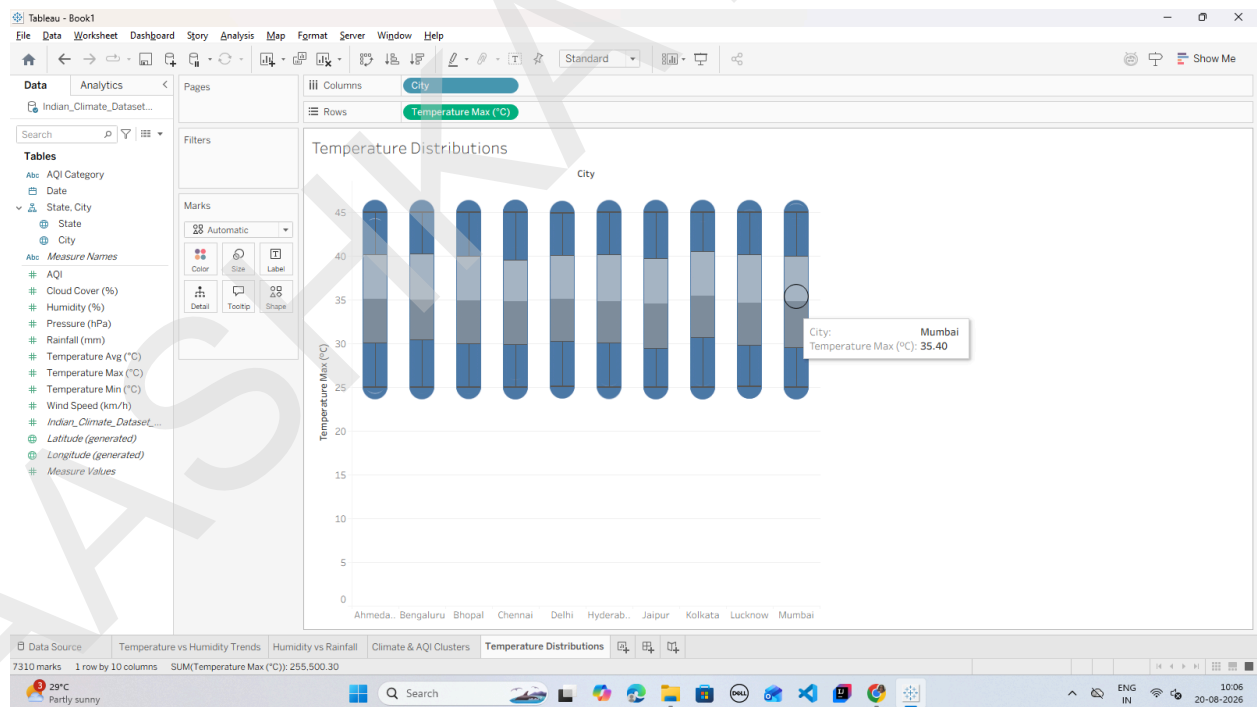
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3. Analyze Cluster Profiles & Save as Dimension



D: Analyze Distributions using Built-in Tools

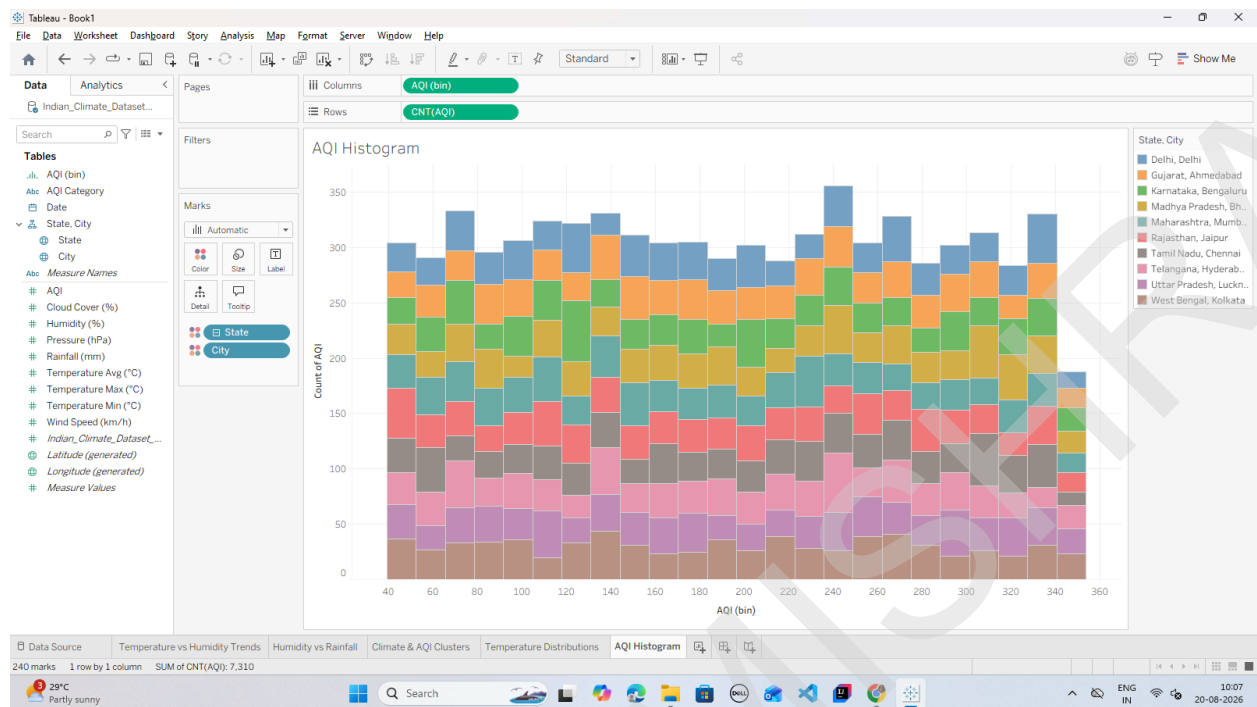
1. Create a Box Plot for Temperature Distributions by City



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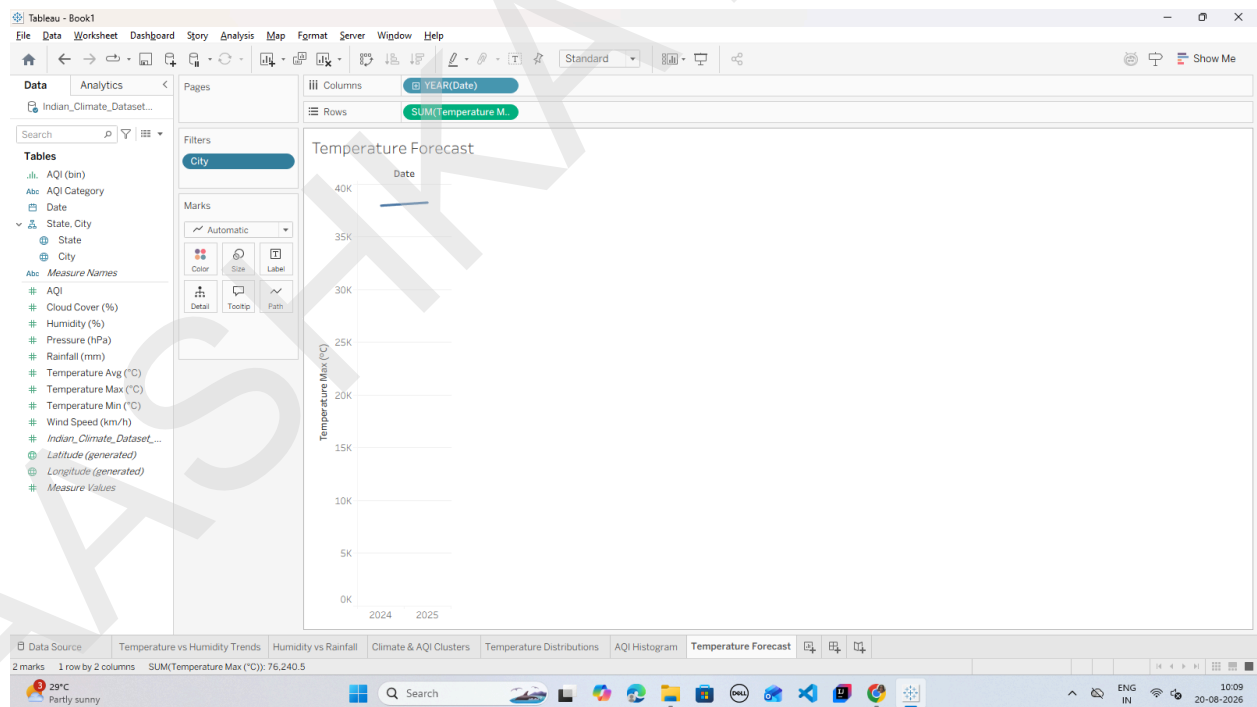
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2. Build an AQI Histogram & Quantile/Percentile View



E: Apply Forecasting Techniques

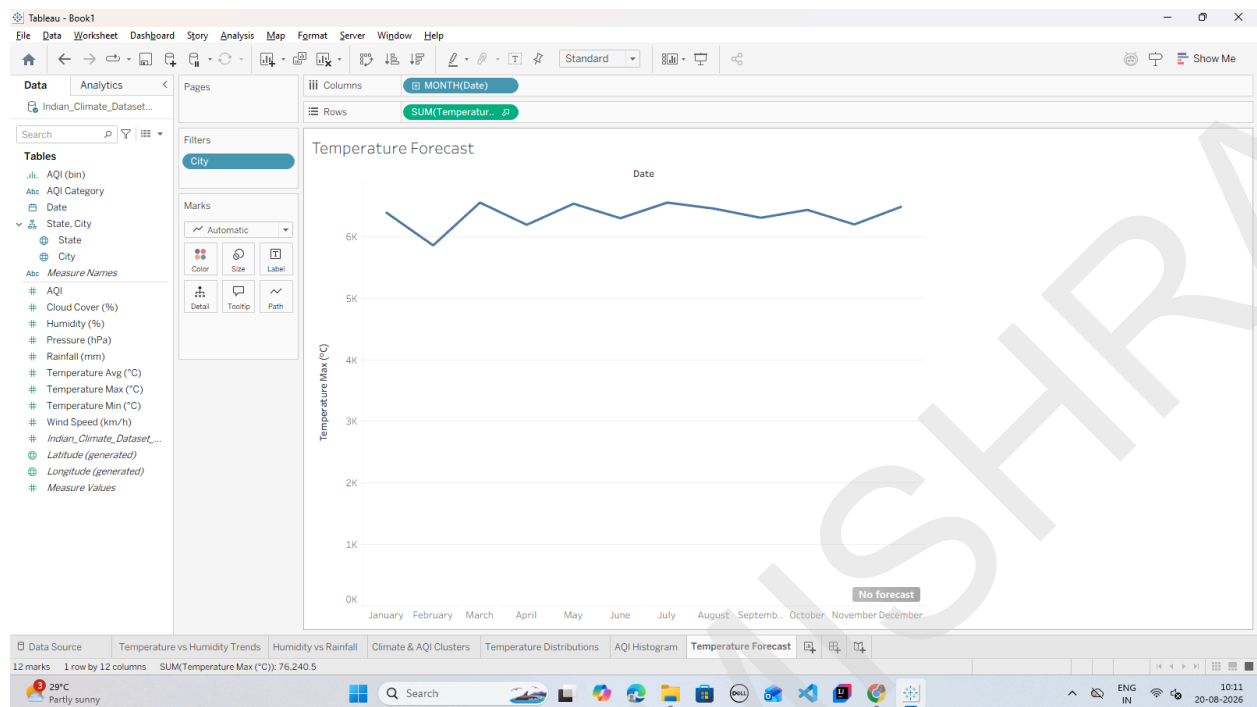
1. Set Up a Time Series Line Chart



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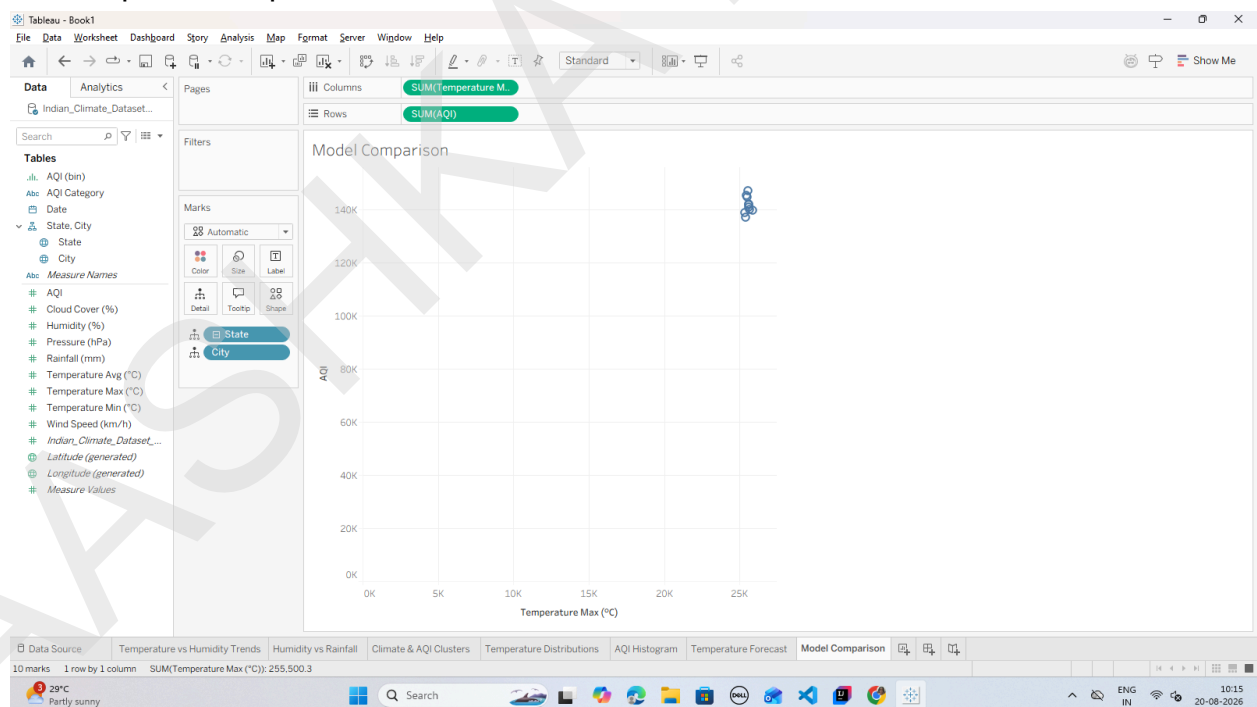
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2. Enable Tableau Forecasting



F: Compare different Statistical Models

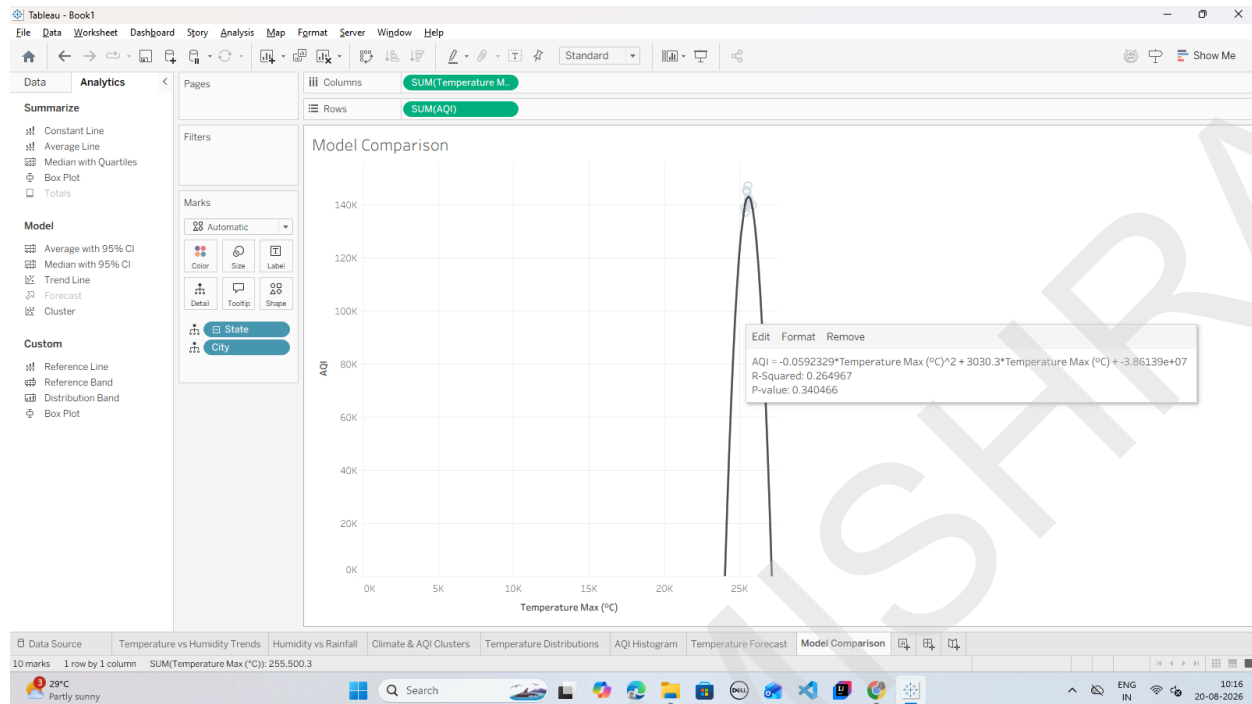
1. Set Up the Comparison View



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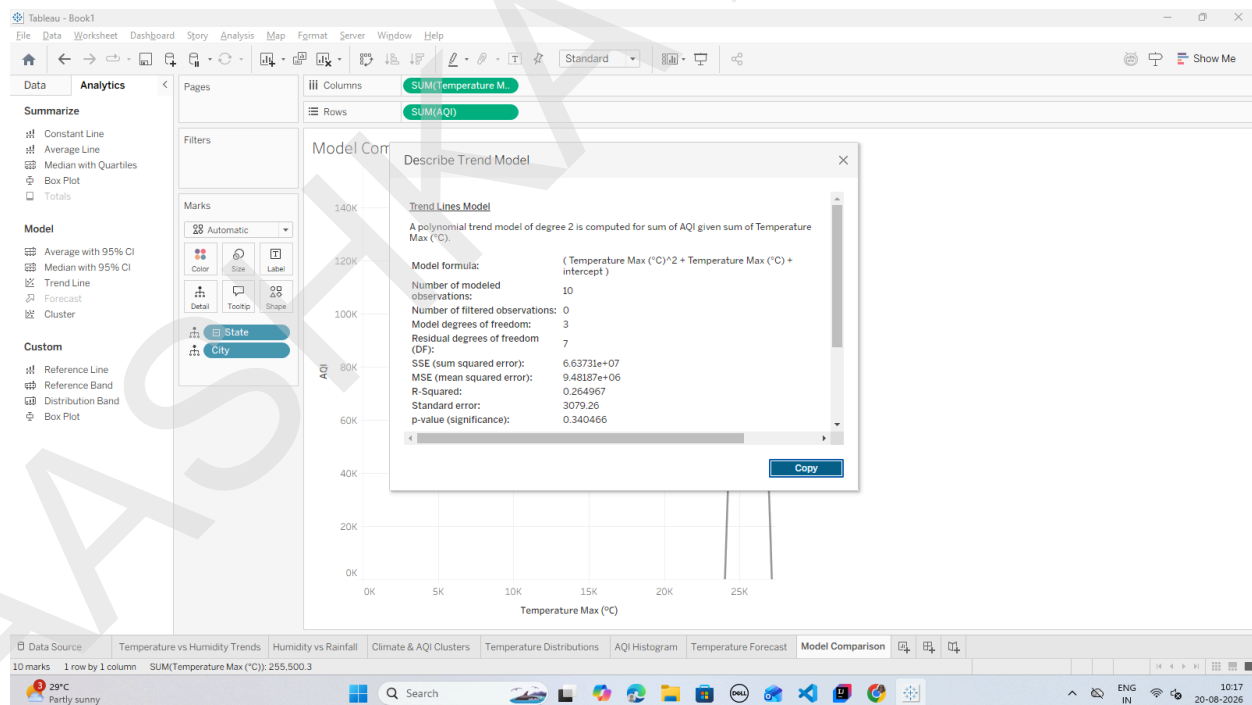
2. Add and Swap Statistical Models



G: Export and Interpret Statistical Results

1. Export the Statistical Model Description

2. Interpret the Key Output Metrics



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